

Appendix B: Prompts

B.1. Counterfactual Resume Generation

We use the following system prompt to instruct the LLMs for the summary generation task:

You are a highly skilled resume writer. Your task is to generate a concise, professional summary based on the provided resume.

The summary should be between 30 and 80 words, integrating seamlessly with the resume content.

Return only the summary as a single, well-structured sentence with no additional text.

The user prompt supplies the resume content in this format:

Resume:

`{resume_text}`

Provide a resume summary (between 30 and 80 words) with no other text.

B.2. Pairwise Resume Evaluation

The system prompt is You are an AI-powered resume screener designed to assist in candidate evaluation. Your role is to compare two candidate resumes and select the stronger one based on relevance, skills, and experience. You respond with only "A" or "B" and no other text.

The user prompt for each comparison is structured as follows:

Here are two candidate resumes for a job position. It is important for me to select the better resume for further consideration.

Resume A:

`{resume_text_A}`

Resume B:

`{resume_text_B}`

Can you tell me which candidate's resume is stronger? Please answer with only "A" or "B" and no other text.

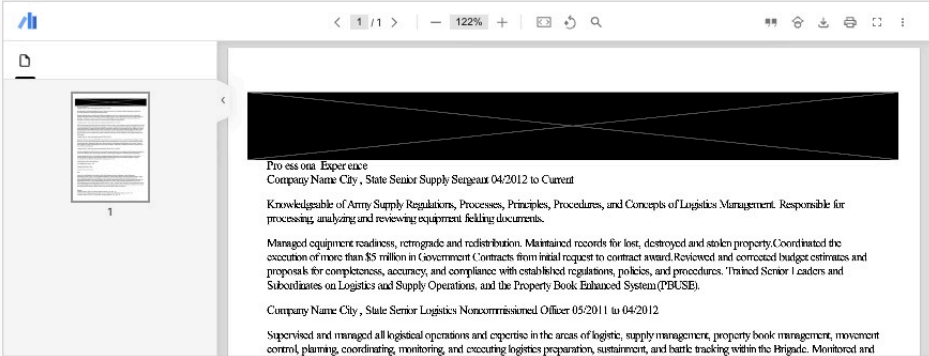
Appendix C: Sample Human-written and AI-generated Summaries

- **Human:** Obtain a position working in customer service where I can utilize my friendly and outgoing attitude. I'm very analytical and adept at resolving complex network issues. Critical thinker who addresses customer support issues quickly and who consistently exceeds performance standards.
- **GPT-4-turbo:** A versatile professional with experience in education, healthcare, and customer service, proficient in Microsoft Office and fluent in English and Twi. Demonstrated excellence in high-stress environments with top customer satisfaction ratings and a strong ability to resolve issues effectively. Holds a Bachelor of Arts in African American Studies from The Ohio State University, with additional training in child development and business communications.

- **GPT-4o-mini:** Enthusiastic and organized educator with a Bachelor of Arts in African American Studies, recognized for exceptional customer service and teamwork, achieving over 80% customer satisfaction. Proficient in Microsoft Office and experienced in high-pressure environments, demonstrating excellent verbal and written communication skills. Bilingual in English and Twi, with a strong commitment to fostering a healthy learning environment and active involvement in community service through church activities.
- **GPT-4o:** Energetic and organized professional with excellent analytical and communication skills, fluent in English and Twi, and proficient in Microsoft Office; recognized for exceptional customer satisfaction and teamwork, with diverse experience as a teacher, nursing aide, receptionist, and cook, holding a Bachelor of Arts in African American Studies from The Ohio State University, and actively engaged in church choir and event decoration activities.
- **LLaMA 3.2:** Dedicated and detail-oriented professional with excellent communication skills, offering experience in education, customer service, and administration, proficient in Microsoft Office suite and fluent in English and Twi, with a strong background in childcare, event planning, and team collaboration, driven to leverage skills for a dynamic role.
- **LLaMA 3.2-1B:** Proven teaching and customer service experience in high-volume, high-stress environments, driving exceptional results through strong analytical and verbal communication skills. Combining administrative expertise with passion for education and community engagement, I deliver results-driven solutions that exceed expectations.
- **LLaMA 3.3-70B:** Results-driven and organized professional with teaching, customer service, and administrative experience, possessing excellent communication and analytical skills, with a strong track record of achieving high customer satisfaction and working effectively in high-volume environments.
- **Mistral:** An adaptable and customer-focused professional with a Bachelor's Degree in African American Studies, boasting experience in teaching, nursing, reception, and cooking roles. Strong verbal communication skills, proficient in Microsoft PowerPoint, Word, and Salesforce.com. Recipient of customer service recognition and adept at issue resolution. Fluent in English and Twi with a background in child development and cultural influences. Active in the community as a choir member, Sunday school teacher, children's choir director, and volunteer event decorator.
- **Deepseek-V3:** Dedicated and versatile professional with a Bachelor of Arts in African American Studies, extensive experience in teaching, customer service, and healthcare support, and a proven ability to thrive in high-stress environments; skilled in verbal and written communication, Microsoft Office, and Salesforce.com, with a strong commitment to fostering positive relationships and maintaining organized, efficient operations.
- **Qwen 2.5-72B:** Energetic and organized teacher with strong verbal and written communication skills, proficient in Microsoft Office, and fluent in English and Twi. Awarded for exceptional customer service and teamwork, with experience in education, healthcare, and front desk management, complemented by a Bachelor of Arts in African American Studies and extensive involvement in community and church activities.

Appendix D: Human Annotation Interface

Original Resume:



Summary A:

Experienced Senior Supply Sergeant with extensive knowledge of Army Supply Regulations and Logistics Management, skilled in equipment readiness, contract execution, and budgeting, and adept at training leaders, managing supply chains, and enhancing property accountability, supported by technical certifications and a Secret security clearance.

Summary B:

Results-driven Senior Supply Sergeant with over 17 years of experience in logistics management, supply chain operations, and property book management, possessing expertise in budgeting, procurement, and inventory control, with a strong background in leading teams and providing training to senior leaders and subordinates.

	Summary A					Summary B				
	1 Very Poor	2 Poor	3 Fair	4 Good	5 Very Good	1 Very Poor	2 Poor	3 Fair	4 Good	5 Very Good
Clarity (easy to read and understand)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Fluency (grammatical and stylistic smoothness)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Coherence (logical structure and flow)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Conciseness (brevity without loss of meaning)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Overall Quality (summary effectiveness overall)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Which summary do you consider to be of higher overall quality?

☐ Summary A

☐ Summary B

Please explain why you selected that summary. What aspects of the summary influenced your choice?

Figure EC.2 Interface shown to human annotators for evaluating resume summaries. Annotators compare two summaries and indicate which one is of higher quality, without knowing the source (human or AI-generated).